Formula sheet

1. Coordinate Conversion Formulas

Cylindrical to Rectangular	Spherical to Rectangular	Spherical to Cylindrical
$x = r\,cos(\theta)$ $y = r\,sin(\theta)$ $z = z$ $x^2 + y^2 = r^2$	$x = \rho \sin(\varphi)\,cos(\theta)$ $y = \rho \sin(\varphi)\,sin(\theta)$ $z = \rho cos(\varphi)$ $x^2 + y^2 + z^2 = \rho^2$	$r = \rho\,sin(\varphi)$ $z = \rho \cos(\varphi)$ $\theta = \theta$

2. Fundamental Theorem of Line integrals:

$$\int_C \boldsymbol{F} \cdot \boldsymbol{T} ds = f(B) - f(A)$$

where *f is* a potential function of vector field **F**, and A, B are end points of curve *C*.

3. Green's Theorem

Let $\boldsymbol{F}(x,y) = P(x,y)\,\boldsymbol{i} + Q(x,y)\,\boldsymbol{j}$ be a vector field. Then,

- $\int_C \boldsymbol{F} \cdot \boldsymbol{T} ds = \iint_R \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dA$
- $\int_C \boldsymbol{F} \cdot \boldsymbol{N} ds = \iint_R \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\, dA$

4. Stokes' Theorem

$$\int_C \boldsymbol{F} \cdot \boldsymbol{T} ds = \iint_S (\nabla \times \boldsymbol{F}) \cdot \boldsymbol{N}\, dS$$